Elementary Catastrophes of Timelike Circular Orbits in Stationary Axisymmetric Spacetimes

A programme paper: the classification is forced to be catastrophe-theoretic,
the cusp locus is counted to be a curve rather than a point,
and the validation ladder and kill tests are stated before the work is done.

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October 2026 **Work in progress — in preparation**

Abstract

The structure of timelike circular orbits around a Kerr–Newman black hole changes qualitatively as the spin a and charge Q are varied: stable and unstable orbits merge and annihilate, and the innermost stable circular orbit (ISCO) is the simplest instance. Individual ISCO radii are classical and tabulated. What is absent is a *classification*: a statement of which degeneracy types can occur, where each occurs in parameter space, and whether the list is complete. This paper states that programme, establishes the two structural results that constrain it, reports the two derivations completed so far, and pre-registers the validation ladder and the conditions under which the programme fails. Three things are established here. First, a structural result: the reduced radial system is conservative rather than dissipative, so it has no asymptotically stable equilibria and no isolated limit cycles; Andronov–Hopf and Bautin bifurcations are therefore excluded as a matter of structure, and the only available transitions are degeneracies of the effective potential as a function of r — that is, the elementary catastrophes A_k. Second, a counting result: with the angular momentum L promoted to a control parameter, the fold A_2 is codimension one, the cusp A_3 is codimension two, and the swallowtail A_4 is codimension three in (a, Q, L); projected to the (a, Q) plane at ISCO-determined L, the cusp locus is a *curve* and not an isolated point. This contradicts the framing under which this project was begun and is the sharpest open question it now carries. Third, the first two rungs of a validation ladder: the Newtonian effective potential is shown to admit no marginally stable circular orbit for any L, establishing that the phenomenon under study is absent without curvature; and the leading post-Newtonian correction reproduces $r_{\mathrm{ISCO}} = 6GM/c^2$ and $L^2 = 12G^2M^2/c^2$. No part of the Kerr–Newman calculation has been performed. The related-work reading is not complete, so the novelty statement here is provisional. We state what is proven, what is counted but unverified, and what has not been attempted.

1 Introduction

1.1 A classical problem without a classification

Circular orbits around black holes are classical. Bardeen, Press and Teukolsky [2] gave the Kerr ISCO as a function of spin in 1972; Chandrasekhar [3] and Misner, Thorne and Wheeler [4] treat the orbital mechanics in full; closed-form expressions for the Kerr–Newman case exist in the literature. Asked for the ISCO radius at a given (a, Q), one looks it up.

What does not exist is a classification. The existing results are pointwise: a radius for each parameter value. They do not answer which *types* of degeneracy occur in the orbital structure, where in parameter space each type occurs, or whether the resulting list is complete. These are not questions about numbers; they are questions about the qualitative organisation of a family of functions, and there is a mature mathematical

theory — Thom's and Arnold's classification of degenerate critical points [7, 1] — that is built precisely to answer them and has not been pointed at this problem systematically.

1.2 The claim

> The transitions in circular-orbit structure of a stationary axisymmetric spacetime are degeneracies of a one-dimensional effective potential, hence elementary catastrophes. Which catastrophes are *available* is a structural question, prior to and independent of where they occur, and it is settled by the dynamical character of the reduced system. We settle it: the reduction is conservative, which excludes every bifurcation requiring an attractor, and leaves exactly the A_k hierarchy. The classification is therefore closed in principle, and the remaining work is to locate its strata.

1.3 Why this is stated before the calculation

A prior project of the author's carried a central claim for several weeks that turned out to be false on inspection, and the correction improved the result. The lesson taken from that episode is that the structural commitments of a programme should be written down, and the conditions for abandoning it fixed, before the hours are spent rather than after. Sections 6 and 7 are that commitment. Section 9 states, in the present tense, what is and is not established.

The structural framing recorded here also corrects a false statement made earlier in this project's own development. The radial reduction was initially described as a *gradient* system, $\dot{x} = -\nabla V$. It is not; it is conservative, and the distinction is not cosmetic, because conservative systems do admit periodic orbits (bounded radial oscillation is exactly such an orbit). Lemma 1 gives the correct argument, which reaches the same exclusion by a different and more defensible route.

1.4 Contributions

1. **The exclusion lemma** (§3). The reduced radial system is conservative, so it has no asymptotically stable equilibria and no isolated limit cycles. Andronov–Hopf and Bautin bifurcations cannot occur, as a matter of structure rather than of observation. The available transitions are exactly the elementary catastrophes.
2. **The codimension count** (§4). In the three-parameter control space (a, Q, L): A_2 has codimension one, A_3 codimension two, A_4 codimension three. Consequently the cusp locus projected to (a, Q) is a curve, and the natural isolated-point object is the swallowtail, not the cusp. This is stated as a counting argument, unverified against the actual Kerr–Newman potential, and it is the programme's sharpest open question.
3. **Validation rungs one and two** (§5). The Newtonian null result, derived: $V''(r_c) = G^4M^4/L^6 > 0$ for all L, so no Newtonian circular orbit is ever marginally stable and the fold does not occur. The post-Newtonian fold, derived: $r_{\mathrm{ISCO}} = 6GM/c^2$ and $L^2 = 12G^2M^2/c^2$, matching the standard Schwarzschild values.
4. **A pre-registered ladder and kill set** (§§6–7). Five rungs, each validated against a known limit before the next is attempted; four kill conditions fixed in advance, including the one under which the programme reduces to a restatement of known results.

1.5 What this paper does not claim

No Kerr–Newman result is claimed. No cusp has been located. The novelty statement in §8 is provisional because the two nearest pieces of prior work have been identified but not read in full, and that reading is a precondition of any novelty claim rather than a formality.

2 The Object

2.1 The reduction

Equatorial motion of a test particle — a body whose own gravitation does not perturb the geometry — requires two coordinates: the areal radius r and the azimuthal angle ϕ. Stationarity and axisymmetry supply two Killing vectors and hence two conserved quantities, the energy E and the angular momentum L, by Noether's theorem. Conservation of L makes ϕ dependent: given L and r, the sweep rate $\dot{\phi}$ is determined. Eliminating it reduces the motion to one degree of freedom,

$$\dot{r}^2 = \mathscr{F}(r; a, Q, L, E), \tag{1}$$

where a dot denotes differentiation with respect to proper time. In the Newtonian limit this is conventionally written $\frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) = E$; in the relativistic case the standard convention is $\dot{r}^2 = E^2 - V(r)$, with

$$V_{\text{Schw}}(r) = \Big(1 - \frac{2M}{r}\Big)\Big(1 + \frac{L^2}{r^2}\Big). \tag{2}$$

The two conventions differ by a constant and a factor and do not affect anything below; we fix $\dot{r}^2 = E^2 - V(r)$ for the relativistic sections.

The function V is not a potential energy. It is the sum of a genuine potential term and the kinetic energy of the azimuthal motion, re-expressed through the conservation law as a function of r alone. The $L^2/2r^2$ term in particular is not a force; it is a conservation constraint, and reading it as a repulsion is a convenience rather than a mechanism.

2.2 Circular orbits as critical points

Definition 1. A *circular orbit* at radius r_* is a critical point of V: $V'(r_*) = 0$. It is *stable* if $V''(r_*) > 0$ and *unstable* if $V''(r_*) < 0$.

The whole problem is now the problem of a family of functions of one variable, indexed by parameters. Where V'' vanishes alongside V', a minimum and a maximum have merged and annihilated; this is the fold A_2, and physically it is the ISCO. Where V''' vanishes in addition, three critical points coalesce; this is the cusp A_3. The hierarchy continues: A_k is the vanishing of $V', V'', \ldots, V^{(k)}$.

3 Structure: the Reduction is Conservative

Lemma 1 (Conservative reduction)**.** *Let $\dot{r}^2 = E^2 - V(r)$ with V smooth. Then the induced second-order system is*

$$\ddot{r} = -\tfrac{1}{2}V'(r), \tag{3}$$

which is conservative, with conserved quantity $H(r, \dot{r}) = \frac{1}{2}\dot{r}^2 + \frac{1}{2}V(r)$. Its linearisation at any equilibrium has spectrum symmetric under $\lambda \mapsto -\lambda$; every equilibrium is a centre or a saddle, and none is asymptotically stable.

Proof. Differentiating $\dot{r}^2 = E^2 - V(r)$ with respect to proper time gives $2\dot{r}\ddot{r} = -V'(r)\dot{r}$, hence $\ddot{r} = -\frac{1}{2}V'(r)$ wherever $\dot{r} \neq 0$, and by continuity everywhere. Writing $v = \dot{r}$, the system is $(\dot{r}, \dot{v}) = (v, -\frac{1}{2}V'(r))$, whose divergence vanishes identically, so phase-space area is preserved. Equivalently $\frac{d}{d\tau}H = v\dot{v} + \frac{1}{2}V'(r)\dot{r} = 0$. The Jacobian at an equilibrium $(r_*, 0)$ is $\left(\begin{smallmatrix} 0 & 1 \\ -\frac{1}{2}V''(r_*) & 0 \end{smallmatrix}\right)$, with eigenvalues $\lambda = \pm\sqrt{-\frac{1}{2}V''(r_*)}$: a pair on the imaginary axis when $V'' > 0$ (a centre) and a real pair of opposite sign when $V'' < 0$ (a saddle). In neither case is there an eigenvalue with negative real part accompanied by no positive one, so no equilibrium is asymptotically stable. □

Corollary 1 (Exclusion of Hopf and Bautin)**.** *No Andronov–Hopf or Bautin (generalised Hopf) bifurcation occurs in this system for any smooth V and any parameter values.*

Proof. An Andronov–Hopf bifurcation requires a conjugate pair of eigenvalues crossing the imaginary axis *transversally* as a parameter varies, together with a nonzero first Lyapunov coefficient, producing an isolated limit cycle from a focus. By Lemma 1 the spectrum is constrained to $\pm\sqrt{-\frac{1}{2}V''}$: when $V'' > 0$ the pair lies on the imaginary axis over an entire open range of parameters rather than crossing it at a point, so the transversality condition fails identically. There is moreover no focus to bifurcate from, since no equilibrium is asymptotically stable. Finally, the conserved quantity H foliates the phase plane into level sets, so any periodic orbit lies in a continuous one-parameter family of periodic orbits and cannot be isolated; limit cycles do not exist. Bautin requires in addition the vanishing of the first Lyapunov coefficient at a Hopf point, and therefore requires a Hopf point, which does not exist here. □

Corollary 2 (The classification is catastrophe-theoretic)**.** *The qualitative transitions available to the circular-orbit structure are exactly the changes in the critical-point set of V as the parameters vary, that is, the elementary catastrophes A_k in Thom's classification.*

Two remarks. First, this is the sense in which catastrophe theory is *forced* here rather than chosen: it is not one lens among several but the only one whose objects can occur. Second, the exclusion is sensitive to the reduction. An analysis retaining the full geodesic phase-plane flow, rather than reducing to the radial effective potential, works with a different object, and degenerate bifurcations unavailable here — including Bogdanov–Takens — can appear there. This is the precise technical boundary between the present programme and the prior work discussed in §8: a difference in reduction, not merely in spacetime.

4 Counting: Where the Strata Live

Proposition 1 (Codimension count)**.** *Regard $V(r;a,Q,L)$ as a family of functions of the state variable r with control parameters $(a,Q,L) \in \mathbb{R}^3$. Assuming the family is generic in the sense of Thom's transversality theorem, the stratum A_k has codimension $k-1$ in control space. In particular:*

- *A_2 (fold, the ISCO) is a surface, codimension one;*
- *A_3 (cusp) is a curve, codimension two;*
- *A_4 (swallowtail) is a set of isolated points, codimension three.*

Projecting to the (a,Q) plane with L fixed by the ISCO condition, the fold becomes a graph over the plane and the cusp locus becomes a curve*, not an isolated point.*

Counting argument. A_k at r_* requires $V'(r_*) = V''(r_*) = \cdots = V^{(k)}(r_*) = 0$, which is k scalar equations in the single unknown r_* together with the controls. Eliminating r_* leaves $k-1$ conditions on the controls. With three controls, A_2 leaves two free parameters, A_3 leaves one, and A_4 leaves zero. For the projected statement: fixing (a,Q), the pair $V' = V'' = 0$ is two equations in the two unknowns (r,L) and generically has isolated solutions, so the ISCO is well-defined pointwise over the plane and determines L. Imposing $V''' = 0$ in addition is a third equation on the same two unknowns, leaving one residual scalar condition on (a,Q), whose zero set is generically one-dimensional. □

This contradicts the framing under which this project was begun, which treated the cusp as an isolated point in the (a,Q) plane to be located. Three readings are possible and the programme does not yet distinguish them:

1. The count is right and the cusp locus is a curve. This would be the *stronger* outcome: a one-dimensional cusp stratum is a generic, structurally stable feature rather than a fine-tuned coincidence, and it is the natural object to contrast with results that locate a cusp only in an extremal limit.
2. The Kerr–Newman family is non-generic in a way that collapses the stratum — for instance through a symmetry or an algebraic degeneracy of the potential — and the cusp is genuinely a point. This would itself be a result worth stating, since a failure of transversality in a physically natural family is not a triviality.
3. The intended object was never A_3 in (a, Q, L) but a singularity of the projected ISCO *curve* regarded as a plane curve, which is a different question with a different count.

Conjecture 1. In the physical region $a^2 + Q^2 \leq M^2$, the A_3 stratum of the Kerr–Newman family is a nonempty one-dimensional subset of the (a, Q) plane, and the A_4 stratum in (a, Q, L) is nonempty and finite.

Resolving this is the first substantive task once the Kerr–Newman potential is in hand, and it is cheap: it is one additional derivative condition on equations that the fold calculation already produces.

5 Established

5.1 Rung one: the Newtonian null result

Proposition 2. *For the Newtonian effective potential $V_{\text{eff}}(r) = -GMr^{-1} + \frac{1}{2}L^2 r^{-2}$, the unique circular orbit is at $r_c = L^2/GM$ and satisfies*

$$V''_{\text{eff}}(r_c) = \frac{G^4 M^4}{L^6} > 0 \tag{4}$$

for every $L \neq 0$. No value of L renders a Newtonian circular orbit marginally stable; the fold does not occur, and there is no ISCO.

Proof. $V'_{\text{eff}} = GMr^{-2} - L^2 r^{-3}$, which vanishes when $GMr = L^2$, giving $r_c = L^2/GM$. Then $V''_{\text{eff}} = -2GMr^{-3} + 3L^2 r^{-4}$, and substituting $r_c^{-1} = GM/L^2$,

$$V''_{\text{eff}}(r_c) = -2GM\frac{G^3M^3}{L^6} + 3L^2\frac{G^4M^4}{L^8} = \frac{(-2+3)\,G^4M^4}{L^6} = \frac{G^4M^4}{L^6}. \tag{5}$$

Every factor appears to an even power, so the expression is strictly positive and never vanishes. □

This is a negative result and it is the programme's baseline. It establishes that the phenomenon under study is produced by curvature and is not a generic feature of central-force orbital mechanics — so that any fold located at a higher rung is attributable to the geometry rather than to the reduction.

5.2 Rung two: the post-Newtonian fold

Proposition 3. *For the leading post-Newtonian potential*

$$V_{\text{eff}}(r) = -\frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2 r^3}, \tag{6}$$

the fold conditions $V'_{\text{eff}} = V''_{\text{eff}} = 0$ have the unique solution

$$r_{\text{ISCO}} = \frac{6GM}{c^2}, \qquad L^2 = \frac{12G^2M^2}{c^2}, \tag{7}$$

reproducing the standard Schwarzschild values $r = 6M$, $L^2 = 12M^2$ in geometric units.

Proof. Differentiating, $V' = GMr^{-2} - L^2 r^{-3} + 3GML^2 c^{-2} r^{-4}$ and $V'' = -2GMr^{-3} + 3L^2 r^{-4} - 12GML^2 c^{-2} r^{-5}$. Multiplying $V' = 0$ by r^4 clears the negative exponents: $GMr^2 - L^2 r + 3GML^2/c^2 = 0$, so that $L^2\big(r - 3GM/c^2\big) = GMr^2$ and

$$L^2 = \frac{GMr^2}{r - 3GM/c^2}. \tag{8}$$

Multiplying $V'' = 0$ by r^5 gives $-2GMr^2 + 3L^2 r - 12GML^2/c^2 = 0$, that is $L^2\big(3r - 12GM/c^2\big) = 2GMr^2$. Substituting (8) and cancelling the common nonzero factor GMr^2,

$$\frac{3r - 12GM/c^2}{r - 3GM/c^2} = 2 \quad \Longrightarrow \quad 3r - \frac{12GM}{c^2} = 2r - \frac{6GM}{c^2} \quad \Longrightarrow \quad r = \frac{6GM}{c^2}. \tag{9}$$

Back-substitution into (8) with $r = 6GM/c^2$ gives $L^2 = GM\,(36G^2M^2/c^4)\big/(3GM/c^2) = 12G^2M^2/c^2$. □

Two by-products. The denominator of (8) diverges at $r = 3GM/c^2$, recovering the photon sphere: no massive particle maintains a circular orbit there at any finite angular momentum. And the correction term carries an explicit c^{-2} and vanishes as $c \to \infty$, returning the potential of rung one — a consistency check built into the structure of the term rather than imposed on it.

6 The Validation Ladder

Each rung is validated against a known limit before the next is attempted. The discipline is that no derivation is trusted until it reproduces an already-published special case, and the ladder exists so that an error is caught at the rung where it enters rather than four rungs later.

1. **Newtonian.** *Complete.* Validation: no ISCO at any L.
2. **Post-Newtonian.** *Complete.* Validation: $r = 6GM/c^2$, $L^2 = 12G^2M^2/c^2$.
3. **Schwarzschild from the metric.** Derive $V = (1 - 2M/r)(1 + L^2/r^2)$ from the metric via the Killing symmetries and the normalisation $g_{\mu\nu}u^\mu u^\nu = -1$. Validation: $r = 6M$ recovered, now from first principles rather than from an imported potential. *Not started.*
4. **Reissner–Nordström.** Charge only. Validation: reduces to rung three at $Q = 0$; ISCO matches published Reissner–Nordström values. Under the method framing this rung is a fully worked second demonstration, not a stepping stone. *Not started.*
5. **Kerr.** Spin only; frame-dragging introduces cross terms. Validation: the Bardeen–Press–Teukolsky ISCO curve [2] reproduced across $0 \le a \le M$, prograde and retrograde. *Not started.*
6. **Kerr–Newman.** Both. Validation: reduces to rung five at $Q = 0$ and rung four at $a = 0$. *Not started.*

7 Pre-registered Kill Conditions

Fixed in advance, so that a negative outcome is a reportable result rather than a retrospective reinterpretation.

K1 — Ladder failure. Any rung fails to reproduce its known limit and the discrepancy is not traced to an identified algebraic error within one working session. The programme stops at the last validated rung and reports the obstruction.

K2 — The counting question. Proposition 1 is tested against the actual Kerr–Newman potential. If the A_3 stratum is empty in the physical region $a^2 + Q^2 \leq M^2$, the result is a precisely bounded negative: the region searched is documented and the emptiness reported. This does not kill the programme, because the method and the completeness claim survive it.

K3 — Completeness is not provable. If the argument that the A_k list is closed cannot be made to a standard a referee would accept — in particular if genericity fails in the Kerr–Newman family in a way that admits strata outside the A_k hierarchy — the paper is demoted from a classification to a survey, and the contribution is correspondingly reduced. This is the condition under which the programme becomes a reorganisation of known results in new language, and it must be reported as such if it obtains.

K4 — Prior work closes the gap. The two identified prior results (§8) are read in full. If either already performs a two-parameter catastrophe classification of timelike circular orbits under the gradient reduction, the novelty claim is withdrawn in full and the work is reported as an independent rederivation.

8 Related Work — Provisional

Two results have been identified as sitting near this programme. Both must be read in full before any novelty claim is made, and that reading has not been done; the statements below are therefore provisional and recorded here so that they can be checked against the papers rather than assumed.

The first, from 2026, is reported to locate a cusp in the transition from stable circular orbit to plunge in Kerr spacetime, with spin varying alone and charge fixed to zero, and with the cusp existing only in an extremal, fine-tuned limit rather than generically across a two-parameter unfolding. If so, the distinguishing features of the present programme are the second parameter and the genericity of the stratum.

The second, in the Klën–Molina lineage, is reported to apply a Bogdanov–Takens analysis to geodesic dynamics in a braneworld spacetime, for null rather than timelike geodesics. The structurally important difference, if the report is accurate, is not the spacetime but the reduction: that analysis retains the full geodesic phase-plane flow, where Corollary 1 does not apply, whereas the present programme works with the conservative radial reduction, where it does. A Bogdanov–Takens point is available in the former and provably unavailable in the latter.

Standard references for the underlying orbital mechanics are [2, 3, 4]; for the catastrophe classification, [7, 1, 6]. The Kerr–Newman metric is taken as given, as an exact solution of the Einstein–Maxwell equations [5], and is not derived here.

9 What Is Proven and What Is Not

Established.

- The reduced radial system is conservative; no equilibrium is asymptotically stable and no isolated limit cycle exists (Lemma 1).
- Andronov–Hopf and Bautin bifurcations are structurally excluded (Corollary 1).
- The Newtonian effective potential admits no marginally stable circular orbit at any L; the fold does not occur without curvature.
- The leading post-Newtonian correction yields $r_{\mathrm{ISCO}} = 6GM/c^2$ and $L^2 = 12G^2M^2/c^2$, and recovers the photon sphere at $r = 3GM/c^2$ as a divergence of the angular momentum.

Counted but not verified.

- The codimension count of Proposition 1, and with it the claim that the cusp locus in (a, Q) is a curve rather than a point. The argument assumes genericity and has not been tested against the Kerr–Newman potential, which does not yet exist in this programme.

Not established, not attempted.

- Any Kerr–Newman result whatsoever. Rungs three through six are not started.
- The completeness of the A_k classification for this family, as distinct from the exclusion of Hopf and Bautin, which is proven. Completeness additionally requires a genericity argument that is not written.
- Any novelty claim. The related-work reading is incomplete (K4).
- Any statement about the astrophysical realisability of the parameter values involved. Charged black holes of astrophysical significance are not claimed; the programme is a statement about the mathematical structure of the Kerr–Newman family.

10 Discussion

What the structural result buys. Corollary 1 is small but it is the load-bearing piece of the programme's framing. Without it, "we use catastrophe theory" is a methodological preference and a reader is entitled to ask why not some other classification. With it, the answer is that no other classification has objects that can occur. It also supplies the clean technical boundary against analyses that retain the full phase-plane flow, which is a sharper distinction than any appeal to a different spacetime.

What the counting result costs. It contradicts the project's founding framing, which is uncomfortable but is the point of doing the count before the calculation. If the cusp locus is one-dimensional, the object to go looking for is the swallowtail, and the three-parameter control space (a, Q, L) is not an optional extension but the correct setting from the start.

Limitations. The programme is at rung two of six. The structural lemma is proven but elementary. The counting argument is a transversality heuristic until a specific potential tests it. The related-work reading is outstanding and gates every novelty statement. The author is self-taught in general relativity and the work has not yet been read by a specialist; arrangements for that review are in progress and the paper will not claim a result before it has occurred.

11 Conclusion

We have stated a classification programme for timelike circular orbits in stationary axisymmetric spacetimes, proven the structural result that forces it to be catastrophe-theoretic, and given a codimension count that relocates the programme's central object from a point to a curve and promotes the swallowtail to the natural isolated stratum. Two rungs of a six-rung validation ladder are complete and both reproduce their known limits. Nothing about Kerr–Newman is claimed. The kill conditions are fixed, including the one under which this becomes a restatement of known results in new language, and that outcome will be reported as such if it obtains.

References

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